Πετράκης Ραφαήλ Ιωάννης ⋆ csd5748

ΗΥ-215: $3^{η}$ Σειρά Ασκήσεων ...

• **Άσκηση $1^{η}$:**

⋆ Αρχικά, γνωρίζουμε: $x(t) = X_0 + \sum_{k=1}^{+\infty} \left[2|X_k|\cos(2\pi k f_0 t + \phi_k)\right]$ και $X_k = |X_k|e^{j\phi_k}$ και $e^{\pm j\pi} = -1$

α) Έχουχε: $x_1(t):\ T_0 = 2,\ X_1 = -1,\ X_3 = 1$ και έτσι: $i = 1,\ k = [0, +\infty),\ X_0 = X_2 = X_{k\geq 4} = 0.$

∗ Είναι: $f_0 = \frac{1}{T_0} = \frac{1}{2}$ και $|X_1| = |-1| = 1,\ \phi_1 = 180° = \pi$ και $|X_3| = |1| = 1,\ \phi_3 = 0° = 0$

∗ Έτσι: $x_1(t) = X_0 + 2|X_1|\cos(2\pi(1)f_0 t + \phi_1) + 2|X_2|\cos(2\pi(2)f_0 t + \phi_2) + 2|X_3|\cos(2\pi(3)f_0 t + \phi_3) + \cdots$

$\implies x_1(t) = 0 + 2\cdot 1\cdot\cos\left(2\pi\cdot 1\cdot\frac{1}{2}\cdot t + \pi\right) + 0 + 2\cdot 1\cdot\cos\left(2\pi\cdot 3\cdot\frac{1}{2}\cdot t + 0\right) + 0\cdots \implies$

∗ Τελικά: $\underline{x_1(t) = 2\cos(\pi t + \pi) + 2\cos(3\pi t)}$

β) Έχουμε: $x_2(t):\ T_0 = 1,\ X_1 = \frac{1}{j},\ X_2 = -j\frac{1}{2},\ X_3 = j\frac{1}{2},\ X_4 = -j\frac{1}{2}$ και έτσι: $i = 2,\ X_0 = 0.$

∗ Είναι: $f_0 = 1,\ |X_1| = |\frac{1}{j}| = |-j| = |0 - 1j| = \sqrt{0^2 + (-1)^2} = 1,\ \phi_1 = 270° = \frac{3\pi}{2} = -\frac{\pi}{2} \in (-\pi, \pi]$ ...

∗ Και: $|X_2| = \frac{1}{2},\ \phi_2 = -\frac{\pi}{2},\ |X_3| = \frac{1}{2},\ \phi_3 = \frac{\pi}{2},\ |X_4| = \frac{1}{2},\ \phi_4 = -\frac{\pi}{2}$

∗ Άρα: $x_2(t) = 0 + 2(1)\cos\left(2\pi(1)(1)t - \frac{\pi}{2}\right) + 2\left(\frac{1}{2}\right)\cos\left(2\pi(2)(1)t - \frac{\pi}{2}\right) + 2\left(\frac{1}{2}\right)\cos\left(2\pi(3)(1)t + \frac{\pi}{2}\right) + 2\left(\frac{1}{2}\right)\cos\left(2\pi(4)(1)t - \frac{\pi}{2}\right)$

∗ Τελικά: $\underline{x_2(t) = 2\cos(2\pi t - \pi/2) + \cos(4\pi t - \pi/2) + \cos(6\pi t + \pi/2) + \cos(8\pi t - \pi/2)}$

γ) Έχουμε: $x_3(t):\ T_0 = 1/2,\ X_2 = 2e^{j\pi/3},\ X_3 = -e^{-j\pi/4}$ και έτσι: $i = 3,\ X_0 = X_1 = 0.$

∗ Είναι: $f_0 = 2,\ |X_2| = 2,\ \phi_2 = \pi/3,\ X_3 = e^{j\pi}\cdot e^{-j\pi/4} = e^{j(\pi - \pi/4)} = e^{j(3\pi/4)} \implies |X_3| = 1,\ \phi_3 = 3\pi/4$

∗ Οπότε: $x_3(t) = 0 + 0 + 2(2)\cos\left(2\pi\cdot 2\cdot 2\cdot t + \frac{\pi}{3}\right) + 2(1)\cos\left(2\pi\cdot 3\cdot 2\cdot t + \frac{3\pi}{4}\right)$

∗ Τελικά: $\underline{x_3(t) = 4\cos(8\pi t + \pi/3) + 2\cos(12\pi t + 3\pi/4)}$

• **Άσκηση $2^{η}$ - Ερώτημα Ι:**

⋆ Ισχύει: $x(t) = A\cos(2\pi f_0 t + \phi) = \frac{A}{2}e^{j\phi}e^{j2\pi f_0 t} + \frac{A}{2}e^{-j\phi}e^{-j2\pi f_0 t}$ (i)

⋆ Και: $x(t) = \sum_{k=1}^{N} A_k\cos(2\pi f_k t + \phi_k) = \sum_{k=1}^{N}\left[\frac{A_k}{2}e^{j\phi_k}e^{j2\pi f_k t} + \frac{A_k}{2}e^{-j\phi_k}e^{-j2\pi f_k t}\right]$ (ii)

∗ Έχουμε: $x(t) = \underbrace{-2}_{a} + 2\cos(2\pi 100t + \pi/3)\underbrace{-\sin(2\pi 250t + \pi/4)}_{b}$

a. Αφού: $\cos(\pi) = -1 \implies 2\cos(\pi) = -2 \implies -2 = 2\cos(\pi)$

b. Και: $-\sin(\theta) = \cos(\theta + \pi/2) \implies -\sin(2\pi 250t + \pi/4) = 1\cdot\cos(2\pi 250t + \pi/4 + \pi/2) = 1\cdot\cos(2\pi 250t + 3\pi/4)$

∗ Έχουμε: $x(t) = 2\cos(\pi) + 2\cos(2\pi 100t + \pi/3) + 1\cos(2\pi 250t + 3\pi/4)$

...

...

∗ Έχουμε: $x(t) = \overbrace{2\cos(\pi)}^{a} + \overbrace{2\cos\left(2\pi 100t + \pi/3\right)}^{b} + \overbrace{1\cos\left(2\pi 250t + 3\pi/4\right)}^{c}$

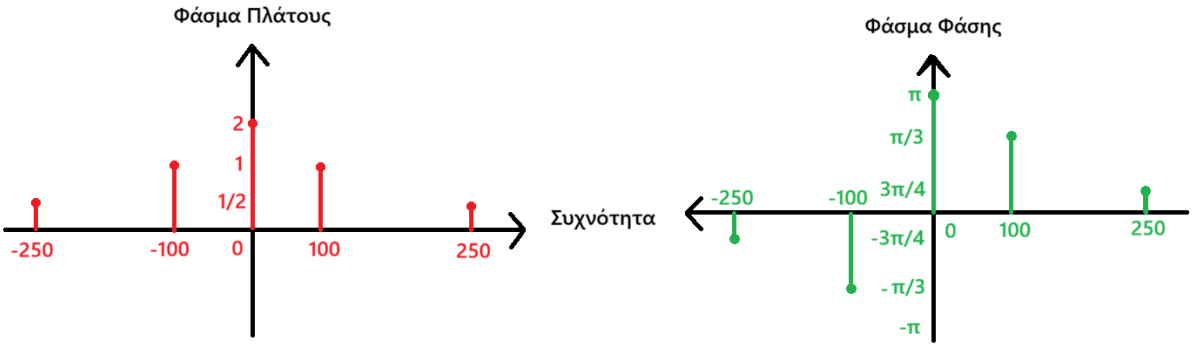
a. $x_0(t) = 2\cos(\pi):\ A_0 = 2,\ f_0 = 0,\ \phi_0 = \pi \overset{(i)}{\Longrightarrow} x_0(t) = 2 \cdot e^{j\pi}$

b. $x_1(t) = 2\cos\left(2\pi 100t + \pi/3\right):\ A_1 = 2,\ f_1 = 100,\ \phi_1 = \pi/3 \overset{(i)}{\Longrightarrow} x_1(t) = 1 \cdot e^{j\pi/3}e^{j2\pi 100t} + 1 \cdot e^{-j\pi/3}e^{-j2\pi 100t}$

c. $x_2(t) = 1\cos\left(2\pi 250t + 3\pi/4\right):\ A_2 = 1,\ f_2 = 250,\ \phi_2 = 3\pi/4 \overset{(i)}{\Longrightarrow} x_2(t) = 0.5 \cdot e^{j3\pi/4}e^{j2\pi 250t} + 0.5 \cdot e^{-j3\pi/4}e^{-j2\pi 250t}$

∗ Τελικά $(ii):\ x(t) = \left[2e^{j\pi}\right] + \left[1 \cdot e^{j\pi/3}e^{j2\pi 100t} + 1 \cdot e^{-j\pi/3}e^{-j2\pi 100t}\right] + \left[0.5 \cdot e^{j3\pi/4}e^{j2\pi 250t} + 0.5 \cdot e^{-j3\pi/4}e^{-j2\pi 250t}\right]$

∗ Και τα σχήματα:



• Άσκηση 2η - Ερώτημα II:

⋆ Γνωρίζουμε: $X(f) = |X(f)|e^{j\angle X(f)}$ και $e^{j\theta} + e^{-j\theta} = 2\cos(\theta)$

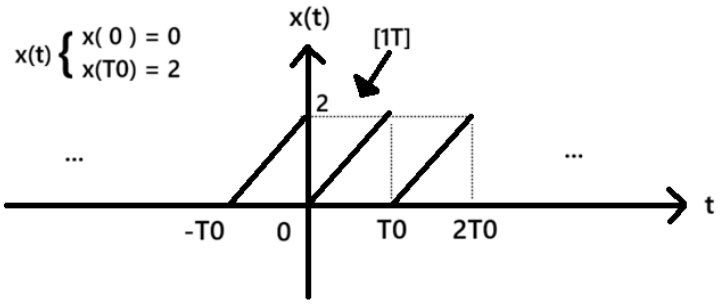
∗ Σύμφωνα με το σχήμα, έχουμε:

– $X(50) = 4e^{-j\pi},\ X(-50) = 4e^{j\pi},\ X(150) = 1.5e^{-j\pi/4},\ X(-150) = 1.5e^{j\pi/4}$

∗ Οπότε είναι: $x(t) = (4e^{-j\pi}e^{j2\pi 50t} + 4e^{j\pi}e^{-j2\pi 50t}) + (1.5e^{-j\pi/4}e^{j2\pi 150t} + 1.5e^{j\pi/4}e^{-j2\pi 150t})$

∗ Ή αλλιώς: $\underline{x(t) = 8\cos(2\pi 50t - \pi) + 3\cos\left(2\pi 150t - \frac{\pi}{4}\right)}$

• Άσκηση 3η - Ερώτημα i:



• **Άσκηση 3η - Ερώτημα ii:**

⋆ Δεδομένα: $X_k = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-j2\pi k f_0 t}dt$ και $X_k = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-j2\pi k f_0 t}dt$

∗ Για k=0: $X_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\cdot 1\ dt = \frac{1}{T_0}\int_0^{T_0} \frac{2}{T_0}t\,dt = \frac{2}{T_0^2}\int_0^{T_0} t\,dt\ \frac{2}{T_0^2}\left[\frac{t^2}{2}\right]_0^{T_0} = \frac{2}{T_0^2}\cdot\frac{T_0^2}{2} = 1$

∗ Για k≠0: $X_k = \frac{1}{T_0}\int_0^{T_0}\left(\frac{2}{T_0}t\right)e^{-j2\pi k f_0 t}dt = \frac{2}{T_0^2}\int_0^{T_0} te^{-j2\pi k f_0 t}dt = \frac{2}{T_0^2}\left[\frac{e^{-j2\pi k f_0 t}}{-j2\pi k f_0}\left(t - \frac{1}{-j2\pi k f_0}\right)\right]_0^{T_0} \implies$

$\implies \frac{2}{T_0^2}\left[\frac{1}{-j2\pi k f_0}\left(T_0 + \frac{1}{j2\pi k f_0}\right) - \frac{1}{-j2\pi k f_0}\left(0 + \frac{1}{j2\pi k f_0}\right)\right] = \ldots = \frac{2}{T_0^2}\cdot\left(\frac{T_0}{-j2\pi k f_0}\right) = \frac{2}{-j2\pi k f_0 T_0} = \frac{2}{-jk2\pi} = \frac{1}{-jk\pi} = \frac{j}{k\pi}$

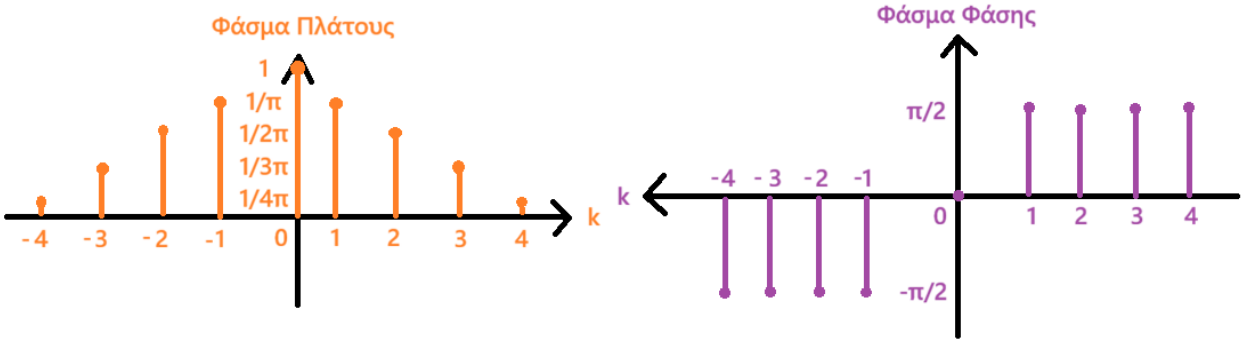
∗ Τελικά: $\underline{X_0 = 1,\ X_k = \frac{j}{k\pi}}$ ή αλλιώς $X_0 = 1,\ X_k = \frac{j}{k\pi} = \frac{1}{k\pi}\cdot j = \frac{1}{k\pi}\cdot\left[\cos\left(\frac{\pi}{2}\right) + j\sin\left(\frac{\pi}{2}\right) = 0 + 1j\right] = \frac{1}{k\pi}e^{j\pi/2}$

• **Άσκηση 3η - Ερώτημα iii:**

∗ Έχουμε: $X_k = \frac{1}{k\pi}j \implies \begin{cases} k = 0: & |X_0| = 1 \text{ και } \phi_0 = 0 \\ k > 0: & |X_k| = 1/k\pi \text{ και } \phi_k = \pi/2 \\ k < 0: & |X_k| = 1/|k|\pi \text{ και } \phi_k = -\pi/2 \end{cases}$

∗ Έτσι: $\begin{cases} k = 0: & |X_0| = 1,\ \phi_0 = 0 \\ k = \pm 1: & |X_1| = |X_{-1}| = 1/1\pi,\ \phi = \{k = 1 : \pi/2,\ k = -1 : -\pi/2\} \\ k = \pm 2: & |X_2| = |X_{-2}| = 1/2\pi,\ \phi = \{k = 2 : \pi/2,\ k = -2 : -\pi/2\} \\ k = \pm 3: & |X_3| = |X_{-3}| = 1/3\pi,\ \phi = \{k = 3 : \pi/2,\ k = -3 : -\pi/2\} \\ k = \pm 4: & |X_4| = |X_{-4}| = 1/4\pi,\ \phi = \{k = 4 : \pi/2,\ k = -4 : -\pi/2\} \end{cases}$

∗ Και το σχήμα:



• **Άσκηση 3η - Ερώτημα iv:**

⋆ Ξανά: $x(t) = X_0 + \sum_{k=1}^{+\infty}\ [2|X_k|\cos(2\pi k f_0 t + \phi_k)]$

∗ Είναι: $\underline{x(t) = 1 + \sum_{k=1}^{\infty} 2\frac{1}{k\pi}\cos\left(2\pi k f_0 t + \frac{\pi}{2}\right)}$

• **Άσκηση 3η - Ερώτημα v:**

∗ Με βάση την ιδιότητα της παραγώγησης, έχουμε:

⋆ $\frac{dx(t)}{dt} = \sum_{k=-\infty}^{+\infty} Y_k e^{j2\pi k f_0 t}$ και $Y_k = j2\pi k f_0 X_k$

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* Είναι: $Y_k = j2\pi k f_0 X_k = j2\pi k f_0 \left(\frac{j}{k\pi}\right) = j \cdot j \cdot 2 \cdot f_0 = j^2 \cdot 2f_0 = -1 \cdot 2 \cdot \frac{1}{T_0} = -\frac{2}{T_0}$ και προφανώς $Y_0 = 0$.

* Τελικά: $\underline{\frac{dx(t)}{dt} = 0 + \sum_{k=-\infty}^{+\infty} \left(-\frac{2}{T_0}\right) e^{j2\pi k f_0 t}}$

★ Άσκηση 4η:

⋆ Δεδομένα: περιοδικό σήμα με περίοδο $T_0 = 2\pi$ και τύπο μίας περιόδου $x(t) = \frac{1}{4}t^2,\ 0 \le t < 2\pi$ ...

* Ισχύει: $x(t) = \sum_{k=-\infty}^{+\infty} X_k e^{j2\pi k f_0 t} = \sum_{k=-\infty}^{+\infty} X_k e^{jk(2\pi\frac{1}{T_0})t} = \sum_{k=-\infty}^{+\infty} X_k e^{jkt}$

1. $X_0:\ X_k = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-j2\pi k f_0 t}dt \overset{k=0}{\Longrightarrow} X_0 = \frac{1}{T_0}\int_0^{T_0} x(t)dt = \frac{1}{2\pi}\int_0^{2\pi}\frac{1}{4}t^2dt = \frac{1}{8\pi}\left[\frac{t^3}{3}\right]_0^{2\pi} = \frac{1}{8\pi}\cdot\frac{8\pi^3}{3} = \frac{\pi^2}{3}$

2. $X_k:\ X_k = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-j2\pi k f_0 t}dt = \frac{1}{2\pi}\int_0^{2\pi}\frac{1}{4}t^2e^{-j(1)kt}dt = \frac{1}{8\pi}\int_0^{2\pi} t^2e^{-jkt}dt = \ \cdots$

[Δεν μπόρεσα να λύσω την υπόλοιπη άσκηση ...]

• Άσκηση 5η:

⋆ Γνωρίζουμε: $P_x = \frac{1}{T_0}\int_{T_0}|x(t)|^2dt$ και $X_k = \frac{1}{T_0}\int_{T_0} x(t)e^{-j2\pi k f_0 t}dt,\ \sin(\theta) = \frac{e^{j\theta}-e^{-j\theta}}{2j}$ και $P_x = \sum_{k=-\infty}^{+\infty}|X_k|^2$

⋆ Δεδομένα: $T_0 = 1 \implies T_0 \in [-0.5,\ 0.5],\ x(t) = \{\ 2 : |t| \le 0.25;\ 0 : 0.25 < t < T_0\ \}$ ...

* Έχουμε: $P_x = \frac{1}{1}\left[\int_{-0.5}^{-0.25} 0^2dt + \int_{-0.25}^{0.25} 2^2dt + \int_{0.25}^{0.5} 0^2dt\right] = \int_{-0.25}^{0.25} 4dt = 4\cdot[t]_{-0.25}^{0.25} = 4\cdot(0.25-(-0.25)) = 4\cdot 0.5 = 2$

* Πρέπει: $P \ge 97\%\cdot P_x \ge 0.97\cdot 2 \ge 1.94$

* Επιπλέον: $X_k = \int_{-0.5}^{-0.25} 0\cdot e^{-j2\pi k f_0 t}dt + \int_{-0.25}^{0.25} 2\cdot e^{-j2\pi k f_0 t}dt + \int_{0.25}^{0.5} 0\cdot e^{-j2\pi k f_0 t}dt = 2\cdot\int_{-0.25}^{0.25}\left(e^{-j2\pi kt}\right)dt$

1. $k = 0:\ X_0 = \int_{-0.25}^{0.25} 2dt = [2t]_{-0.25}^{0.25} = 2(0.25)2(0.25) = 2\cdot 0.5 = 1$

2. $k \ne 0:\ X_k = \int_{-0.25}^{0.25} 2e^{-j2\pi kt}dt = 2\left[\frac{e^{-j2\pi kt}}{-j2\pi k}\right]_{-0.25}^{0.25} = \frac{2}{-j2\pi k}\left(e^{-j2\pi k(0.25)} - e^{-j2\pi k(-0.25)}\right) = \frac{2}{-j2\pi k}\left(e^{-j\frac{\pi k}{2}} - e^{j\frac{\pi k}{2}}\right) \implies$

$\implies \frac{2}{-j2\pi k}\left(-2j\sin\left(\frac{\pi k}{2}\right)\right) = \frac{2}{\pi k}\cdot\sin\left(\frac{\pi k}{2}\right)$

* Είναι: $P = \sum_{k=-\infty}^{+\infty}|X_k|^2 \ge 1.94 \Rightarrow P = \{\ |X_k|\ |\ k \in [-7,\ 7]$ ακέραιος $\} \ge 1.94 \Rightarrow P \approx 1.94959 \ge 1.94$ που ισχύει.

* Τελικά, χρειαζόμαστε $\underline{15\text{ συντελεστές}}$ εκθετικής σειράς Fourier (από τους οποίους οι 6 είναι μηδενικοί) ...

• Άσκηση 6η:

α) Έχουμε: $X_a(f) = \frac{j4\pi f}{1+4\pi^2 f^2}$

* Παρανομαστής: $1 + 4\pi^2 f^2 = 1 - (-4\pi^2 f^2) = (1)^2 - (j2\pi f)^2 = (1 - j2\pi f)(1 + j2\pi f)$

...

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* Άρα: $X_a(f) = \frac{j4\pi f}{(1-j2\pi f)(1+j2\pi f)} = \frac{A}{1-j2\pi f} + \frac{B}{1+j2\pi f} = \cdots$

* Και: $j4\pi f = A(1+j2\pi f) + B(1-j2\pi f) \Rightarrow (0) + (2)j2\pi f = (A+B) + (A-B)j2\pi f \Rightarrow \left.\begin{matrix} A+B=0 \\ A-B=2 \end{matrix}\right\} A=1,\ B=-1$

* Οπότε: $X_a(f) = \frac{1}{1-j2\pi f} - \frac{1}{1+j2\pi f}$

⋆ Ιδιότητες: $\begin{cases} e^{-at}u(t) \overset{F}{\longleftrightarrow} \frac{1}{a+j2\pi f} \overset{a=1}{\Longrightarrow} e^{-t}u(t) \overset{F}{\longleftrightarrow} \frac{1}{1+j2\pi f} \\ e^{at}u(-t) \underset{F}{\longleftrightarrow} \frac{1}{a-j2\pi f} \underset{a=1}{\Longrightarrow} e^{t}u(-t) \underset{F}{\longleftrightarrow} \frac{1}{1-j2\pi f} \end{cases}$

* Τελικά: $\boxed{x_a(t) = e^{t}u(-t) - e^{-t}u(t)}$

β) Έχουμε: $X_b(f) = \frac{4\sin^2(2\pi f)}{(2\pi f)^2}$

⋆ Γνωρίζουμε: $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$

* Είναι: $X_b(f) = 4 \cdot \left(\frac{\sin(2\pi f)}{2\pi f}\right)^2 = 4 \cdot \left(\frac{\sin(\pi \cdot 2f)}{\pi \cdot 2f}\right)^2 = 4 \cdot \text{sinc}^2(2f)$

* Ιδιότητα: $A\text{tri}\left(\frac{t}{T}\right) \overset{F}{\longleftrightarrow} AT\text{sinc}^2(fT) \overset{A=T=2}{\Longrightarrow} 2\text{tri}\left(\frac{t}{2}\right) \overset{F}{\longleftrightarrow} 4\text{sinc}^2(2f)$

* Τελικά: $\boxed{x_b(t) = 2\text{tri}\left(\frac{t}{2}\right)}$

• Άσκηση 7η:

⋆ Δεδομένα: $X(f) = \{1/(2\pi^2) :\ |f| < 1 \mid 0 :\ \text{αλλού}\}$ και $y(t) = d^2x(t)/dt^2 + 4\pi^2 x(t)$ και $\int_{-\infty}^{\infty} |y(t)|^2 dt = 64/15$ ...

⋆ Γνωρίζουμε: $\int_{-\infty}^{\infty} |y(t)|^2 dt = \int_{-\infty}^{\infty} |Y(f)|^2 df$ και $\frac{d^n x(t)}{dt^n} \overset{F}{\longleftrightarrow} (j2\pi f)^n X(f) \overset{n=2}{\Longrightarrow} \frac{d^2x(t)}{dt^2} \overset{F}{\longleftrightarrow} (j2\pi f)^2 X(f) = -4\pi^2 f^2 X(f)$

* Θα έχουμε: $Y(f) = -4\pi^2 f^2 X(f) + 4\pi^2 X(f) = 4\pi^2(1-f^2)X(f)$

* Για $-1 < f < 1$ είναι: $Y(f) = 4\pi^2(1-f^2) \cdot \frac{1}{2\pi^2} = 2(1-f^2)$ και αλλού: $Y(f) = 0$

* Έτσι: $\int_{-\infty}^{\infty} |y(t)|^2 dt = \int_{-1}^{1} |2(1-f^2)|^2 df = 4\int_{-1}^{1} (1-f^2)^2 df = 4\int_{-1}^{1} (1-2f^2+f^4) df = 4\left[f - \frac{2f^3}{3} + \frac{f^5}{5}\right]_{-1}^{1} \Longrightarrow$

$\Longrightarrow 4\left(\left(1 - \frac{2}{3} + \frac{1}{5}\right) - \left(-1 - \frac{2(-1)}{3} + \frac{-1}{5}\right)\right) = 4\left(\left(1 - \frac{2}{3} + \frac{1}{5}\right) - \left(-1 + \frac{2}{3} - \frac{1}{5}\right)\right) = 4\left(1 - \frac{2}{3} + \frac{1}{5} + 1 - \frac{2}{3} + \frac{1}{5}\right) \Longrightarrow$

$\Longrightarrow 4\left(2 - \frac{4}{3} + \frac{2}{5}\right) = 4\left(\frac{30}{15} - \frac{20}{15} + \frac{6}{15}\right) = 4\left(\frac{16}{15}\right) = \frac{64}{15}$

* Τελικά, ισχύει: $\boxed{\int_{-\infty}^{\infty} |y(t)|^2 dt = 64/15}$

ΤΕΛΟΣ
(ευχαριστώ)

LaTeX